MISSION DESIGN

An Interdisciplinary Systems-Design Challenge Framed Through Science Fiction

Designed and facilitated by Gage Harter | PSU STEM Achievers | Grades 6-9

LENGTH	SESSIONS	COHORT	TEAMS	FORMAT
10 Days	**70 Minutes**	**~65 Students**	**8**	**Public Exhibition**

CURRICULUM OVERVIEW

Mission Design is a collaborative systems challenge in which student teams create a fictional world, develop an adapted species and civilization, and respond to a team-specific disruption called the Mission Turn. Students must connect environmental science, biological adaptation, engineering, technology, transportation, governance, and community values. There is no single correct solution; teams are evaluated on system coherence, use of evidence, testing and revision, collaboration, and communication.

10-DAY LEARNING ARC

DAYS	PHASE	STUDENT WORK
1-3	**Build the System**	Form team identity, investigate terrain, climate, resources, adaptation, body structure, and movement through role-based expertise.
4	**High Council**	Debate and decide settlement, survival, transportation, robotics, technology level, governance, and shared values.
5	**Mission Turn**	Analyze a tailored discovery, opportunity, and set of pressures that emerge from the team's own world and decisions.
6-10	**Design, Test, Present**	Plan, model, build, code, test, gather feedback, revise, and present a coherent response in a public exhibition.

SCIENTIFIC & ENGINEERING PRACTICES

MODEL SYSTEMS	ARGUE FROM EVIDENCE	DESIGN & ITERATE	COMMUNICATE INFORMATION
Represent relationships among environment, organisms, resources, technology, and society.	Use research, constraints, and test results to justify decisions and acknowledge trade-offs.	Define problems, prototype, test, document failure, gather feedback, and improve.	Combine scientific reasoning, technical evidence, visuals, and shared speaking.

ASSESSMENT CRITERIA

SYSTEM COHERENCE	EVIDENCE & RATIONALE	TESTING & REVISION	COLLABORATION & COMMUNICATION
Do the world, species, civilization, and solution work together logically?	Are decisions supported by science, constraints, observations, or research?	Can the team show how feedback or failure improved the design?	Does each member contribute, explain, and respond to questions clearly?

ACCESS, EVIDENCE & TOOLS

ACCESS & DIFFERENTIATION	EVIDENCE OF LEARNING	TOOLS & MEDIA
Choice menus, expert roles, visual organizers, scripted discussion supports, multiple product formats, and scaffolded technical pathways.	Research forms, council records, mission response plans, sketches, code, prototypes, test notes, revision evidence, and final presentation.	Tinkercad, Delightex VR/AR, micro:bit, Sphero RVR, coding, digital modeling, robotics, physical prototyping, and presentation media.

Each 70-minute session balances concise instruction, community-building, sustained design time, and a closing reflection or progress check.